

Address Book

Project — Validation Specification

1. Create Contact

The user is prompted to enter contact details repeatedly until all inputs pass validation.

1.1 Name Validation

1. Name must contain only alphabets (uppercase or lowercase), spaces, and dots (.). No other characters are accepted.
2. Name must not start with white-space characters (space, \n, \t).
3. Duplicate names are accepted.
4. Multiple consecutive spaces must be replaced with a single space.

1.2 Phone Number Validation

5. Must contain exactly 10 digits.
6. Must start with 6, 7, 8, or 9.
7. Only digits (0–9) are accepted.
8. Duplicate phone numbers are not accepted.

1.3 Email Validation

9. Must follow the format: mail_name@domain.com
10. mail_name must contain only lowercase alphabets and digits. No other characters are accepted.
11. mail_name is mandatory — a bare @domain.com is not accepted.
12. Domain must contain only lowercase alphabets. Digits and special characters are not accepted.
13. .com is mandatory.
14. Only one @ symbol and one dot (.) are accepted.
15. No characters are accepted after .com
16. Duplicate email addresses are not accepted.

2. Search Contact

The user can search for a contact by Name, Phone Number, or Email.

2.1 Search by Name

- 17. The entered name must pass all Name Validations.
- 18. If the entered name matches multiple records, all matching contacts are displayed.

2.2 Search by Phone Number or Email

- 19. The entered phone number or email must pass all respective validations.
- 20. Only one matching record will exist, since duplicates are disallowed during creation.

3. Edit Contact

The user first searches for the contact to edit (by Name, Phone Number, or Email). If found, the user selects which field to edit.

3.1 Search Phase

- 21. Validate the entered search input.
- 22. If not found → display a "NOT FOUND" message.
- 23. If found → prompt the user to select which field to edit (Name, Phone Number, or Email).

3.2 Edit Name

- 24. Validate the new name.
- 25. If it passes all validations, update the name.

3.3 Edit Phone Number

- 26. Validate the new phone number.
- 27. The new number must not already exist in the existing records.
- 28. If it passes all validations, update the phone number.

3.4 Edit Email

- 29. Validate the new email address.
- 30. The new email must not already exist in the existing records.
- 31. If it passes all validations, update the email address.

4. Delete Contact

The user first searches for the contact to delete (by Name, Phone Number, or Email). Deletion requires explicit confirmation.

4.1 Delete by Name

- 32. The entered name must pass all Name Validations.
- 33. If the name matches multiple records, ask the user to enter the index of the specific contact to delete.
- 34. Ask for confirmation before deleting. Do not delete if confirmation is not given.
- 35. If the entered name does not match any record, no confirmation prompt or deletion should occur.

4.2 Delete by Phone Number or Email

- 36. The entered phone number or email must pass all respective validations.
- 37. If a match is found, ask for confirmation before deleting. Do not delete without confirmation.
- 38. If no match is found, no confirmation prompt or deletion should occur.

5. List Contacts

All contact records are displayed on screen in the following tabular format:

Index	Name	Phone Number	Email
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6. Save and Exit

- 39. All contact data is saved to a .txt file (to persist data across program restarts).
- 40. After saving, the program exits.

7. Initialization

- 41. On startup, all previously saved data is loaded from the .txt file into the program.
- 42. After loading, the program displays the Options Menu and begins normal operation.

 *The program must not terminate until the user explicitly selects "Save and Exit". After every operation, the program must ask the user what they would like to do next.*